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By electronic submission

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Re: European Patent Application No. 22813979.6

Haleon UK IP LIMITED

Dear Sirs,

With reference to the Communication pursuant to Rules 161 (1) and 162 EPC dated 18 June 2024, we file herewith our response to the Written Opinion of the International Searching Authority that was drawn up by the EPO during the International phase of the present application.

Since the application contains only 14 claims presently on file, no excess claim fees are payable.

Comments

Novelty (Article 54 EPC)

In the IPRP, the Examiner has objected that Claim 1 allegedly lacks novelty over the disclosures of cited documents D1 (US4694844) and D2 (DE19506597). We respectfully submit that the Examiner is incorrect to raise these objections for at least the following reasons.

Claim 1 recites that the toothbrush comprises "*at least one deformable element (3), where the deformable element (3) has a first flexible arm (31) arranged relatively towards the cleaning zone (2) and a second flexible arm (32) arranged relatively away from the cleaning zone (2)…**where the arms (31, 32) form a wedge being connected together at one end (33) of the deformable element and distanced from each other at the other end (34) of the deformable element**, where the first*

and second arms (31, 32) are linked by at least two bridges (35, 35', 35") (emphasis added)".

Neither D1 nor D2 disclose such a toothbrush. As shown in e.g. Figures 1 and 2 of D1, to the extent that cover plate 16 and plate 20 are considered flexible arms of a deformable element and flexible, finger-like projections are considered to be bridges linking these arms, it can be seen that plate 20 and cover plate 16 are ***not*** arranged to form a wedge being connected at one end and distanced from each other at the other end (as shown in e.g. Figure 1 of the present application). In clear contrast, plates 20 and 16 in Figure 1 and 2 of D1 are completely parallel and spaced equally at both ends. For at least these reasons, the subject matter of Claim 1 must be considered novel over the disclosure of D1.

Similarly, as shown in figure 7 of D2, the disclosed toothbrush heads do not contain any elements that could be considered to be flexible arms of a deformable element that form a wedge with the arms connected at one end and distanced at another end, where "at least two bridges" separate the arms. To the extent that the portions of bristle carrier 1 separating voids 34 in Figure 7 of D2 could be considered bridges, there are no flexible arms present that these bridges connect since handle 33 is not disclosed as being flexible. There is certainly no disclosure of any structures that could be considered flexible arms connected to form a wedge as described above. For at least these reasons, Claim 1 of the present application is novel over D2.

It is thus considered that the requirements of Article 54 EPC are fulfilled and that the Examiner's objections should be withdrawn.

Inventive Step (Article 56 EPC)

We submit that the subject matter of Claim 1 of the present application involves an inventive step in view of the teachings of cited documents D1 and D2 for at least the following reasons.

As discussed in the paragraph bridging pages 1 and 2 of the application as filed, in the toothbrush of the present invention, the flexible arms are arranged as a wedge and are as such connected at one end of the arms and separated at the other end of the arms. The bridges connecting the arms thus increase in length from one end of the wedge to the other. As discussed in the application as filed, this arrangement mimics the structure of a fish fin and has been found by the present inventors to be important for achieving the effects underlying the present invention. As discussed, the arrangement of the arms and bridges cause the base and tip of the wedge to deform towards an applied force

when the toothbrush head is used for cleaning allowing the cleaning zone to adopt the curvature of a surface of a dentition during cleaning.

Starting from the teaching of either D1 or D2, it would not be obvious for a skilled person to modify the toothbrush heads disclosed therein to comprise flexible arms and bridges arranged in a wedge structure in order to achieve the advantages discussed above underlying the present invention. Neither D1 nor D2 teach or direct the skilled person towards a wedge arrangement of the arms and bridges for any reason whatsoever, and certainly not to achieve the advantages discussed above. For at least these reasons, the claimed subject matter must be considered to involve an inventive step.

Additionally, it would be counterintuitive for a skilled person to modify the brushes disclosed in D1 or D2 to be within the scope of the claims of the present application. As shown in Figure 1 of D1 and Figure 7 of D2, the toothbrushes disclosed in these documents operate via specific mechanisms where the bristles are urged downwards on carrying out cleaning. It would not be obvious to a skilled person whether or not modifying the brushes disclosed in D1 or D2 to be within the scope of the claims of the present application would interfere with these specific mechanisms. Accordingly, absent a specific motivation for doing so, it would not be obvious for a skilled person to modify the disclosed brushes to be within the scope of Claim 1 of the present application without the benefit of impermissible hindsight.

For at least these reasons, the claimed subject matter must be considered to be not obvious for a skilled person and to involve an inventive step. The requirements of Article 56 EPC are thus fulfilled.

Other matters

We wish to defer making any amendments to the description, such as may be necessary, until such time as the Examiner has acknowledged that the claims are in an allowable form.

Conclusion

In light of the comments that are filed herewith, we believe that the present set of claims is allowable. We therefore look forward to an acknowledgement of this from the Examiner. If any minor matter is outstanding, however, then we request that the Examiner telephones the undersigned representative and/or issues a Communication under Article 94 (3) EPC, with a period set for response.

Purely by way of precaution, we hereby request Oral Proceedings in the event that the Examining Division forms an intention to refuse the present application.

As a further precaution, it is also hereby expressly stated that any amendment to the present application, including any deletion, does not amount to an abandonment of any of the subject matter in the application as filed, and that the right to restore deleted subject matter and/or to file one or more divisional applications in respect of any of the subject matter within the application as filed, is maintained.

Yours faithfully,

Charles Dempster

Authorised Representative